

Hoang Vu

647-540-7166 | Email | LinkedIn | Github

EDUCATION

York University | GPA 7.71/9

Honours Bachelor of Science in Computer Science

Sep 2024 – Apr 2026

Toronto, ON

Sheridan College | GPA 3.78/4

Advanced Diploma in Software Development and Network Engineering

Sep 2020 – Dec 2023

Oakville, ON

PROJECTS

MindForge | *FastAPI, React, TypeScript, Supabase, OpenAI, Groq, Hugging Face, Ollama CLOUD*

- Built microlearning platform auto-generating personalized lessons with quizzes in **3 minutes**
- Designed **autonomous 7-agent system** (orchestrator, researcher, content writer, TTS, quiz generator, QA, curator) with retry logic and fallback providers **reducing content creation costs by 80%**

Academic Scheduler | *FastAPI, React, TypeScript, Supabase, OpenAI, PaddlePaddle, LightGBM*

- Built intelligent scheduler reducing assignment deadline clustering by **40%** across **50 student schedules** compared to manual planning baseline
- Integrated PaddleOCR-VL for syllabus parsing and trained LightGBM on **100+ course schedules** to predict workload distribution patterns

EXPERIENCE

Software Developer

SIRT Centre

May 2023 – Aug 2023

Toronto, ON

- Designed and implemented multiplayer menu system in C++ featuring character selection, persistent profile management and real-time session initialization for **10+ concurrent players**
- Decreased crash frequency by **resolving 5+ memory leaks and race conditions**, using systematic debugging, thread-safety analysis, and unit tests

Software Developer

Cloud DX

May 2022 – Dec 2022

Oakville, ON

- Built voice controlled interaction system for HoloLens2, showcased at **ANA Avatar XPRIIZE 2022** with **50+** live demonstrations
- Optimized Arduino-HoloLens communication by refactoring C++/C# data processing, **reducing latency by 30%** for real-time vital sign display
- Implemented gesture-based interaction for HoloLens2, **improving task completion speed by 20%** through usability testing

Software Developer

Medtech Canada

Jan 2022 – Apr 2022

Oakville, ON

- Migrated Unity VR clinic simulation from deprecated XR Rig to XR Origin architecture resulting in **40% faster scene load times**
- Rapidly resolved a critical rendering bug within **48 hours**, restoring accurate motion tracking and eliminating all hand-tracking glitches

TECHNICAL SKILLS

Languages: Java, C#, Python, C++, TypeScript

Frontend: React, Bootstrap, jQuery

Backend: FastAPI, Spring Boot, ASP.NET, Node, Express

Database: PostgreSQL, MongoDB, Supabase

Tools: Git, VS Code, Visual Studio, Eclipse